A greenhouse you can see

Detailed design atlas

41 sections. One coherent Melooloo facility. Realistic materials, crop systems, service rooms and postharvest operations.

200 m²

BP greenhouse module: 10 × 20 m

420

Planting anchors: 14 rows × 30

41 + 1

Every section plus overall view

Photo-realistic finish studies derived from the Blender layout. The model and dimensioned plans retain the editable concept geometry; image details may vary.

Nursery → production → QC → packing → central cold room → dispatch. Utilities, staff spaces and optional visitor/research areas are documented individually.

Three scales. Clear design assumptions.

PRODUCTION MODULE

BP: 10 × 20 m; eave 3 m; ridge 4.5 m; trellis 2.5 m; DFT support 0.8 m. Fourteen rows × thirty anchors = 420. The seven paired rows share six proposed 0.8 m aisles and 0.85 m edge circulation.

INTERNAL ZONING

Proposed front 2.5 m, crop 15 m and rear 2.5 m bands. The crop band is 150 m² gross including paths. It does not establish the BP's 140 m² net planting area. G09 and G11 are envelope/sensor views, not extra rooms.

WET AND DRY SERVICES

Two working reservoirs plus two smaller A/B concentrate containers illustrate role separation. Tank BOM, dosing, flow, pump duties, root-zone volume and climate performance need reconciliation. Dry controls remain separated from wet handling.

REALISTIC APPEARANCE

UV film with insect mesh is the selected visual direction. Galvanized frames, stainless worktops, cream washable surfaces, forest green equipment accents and practical gray floors connect all images.

SHARED SITE ROOMS

Nursery, water treatment, stores, workshop, staff rooms and logistics are separate shared functions. Their sizes and furniture are visualization proposals. The BP office stays at 9 m² per location. Section areas must not be summed into an approved site plan.

CENTRAL POSTHARVEST HUB

One cold room at HQ in Y0. S16–S25 detail receiving, QC, hold, packing, Gift Edition assembly, dry packaging storage, cold holding, dispatch, crate wash and waste. Gift boxes show three fruit per box.

EXTERNAL AND OPTIONAL AREAS

Generator is outdoors; its 3 × 4 m pad is not a separation distance. S26 is a ground loading/lift vignette. S27 is a circulation section between greenhouse edges. Visitor reception and research rooms are optional, without added funded commercial capacity.

HOW TO USE THIS ATLAS

Use the images for atmosphere, finishes and equipment appearance. Use the Blender file and plans for dimensioned layout discussion. The proposed rooms require actual site dimensions, roof loads, vendor access, hygiene, drainage and egress review before construction.

G01 / Modul produksi

Pintu masuk dan vestibule higiene

Visual allocation 2.5 × 2.5 m (concept) | Usulan pemisahan fungsi

Realistic finish study based on Blender concept; small image details may vary.

FUNCTION

Transisi operator dari sirkulasi site ke area budidaya. Pintu GH ada pada BP; vestibule belum dirinci.

LAYOUT

Dua portal berurutan; dispenser dan log pada sisi kiri; threshold rata, lintasan trolley tetap bebas.

EQUIPMENT

Pintu berurutan/insect barrier, titik higiene tangan/alas kaki sesuai SOP, log masuk.

CONNECTIONS

S12 locker/higiene → G01 → G05 jalur kerja.

SERVICES

Lampu tertutup; dispenser tangan; wastafel utama di S12.

FINISHES

Matte light-gray dry floor; threshold rata; furniture mudah dibersihkan. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

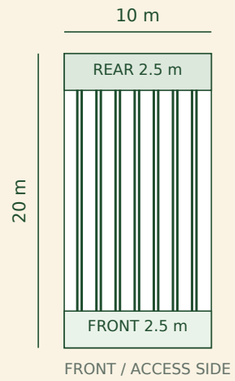
DESIGN REVIEW

Pintu konsep 1,1 m; verifikasi egress dan trolley. Hindari vestibule yang membuang akses panen.

SOURCE / STATUS

Bab V [0878-0884]; vestibule = usulan. Wastafel dapat berada di area bersama.

DIMENSIONED CONCEPT



Same 10 × 20 m module for G01–G12.

14 rows × 30 planting anchors.

Front / crop / rear: 2.5 / 15 / 2.5 m.

150 m² gross crop band; 140 m² net unresolved.

Overlays and vignettes are not additive areas.

G02 / Modul produksi

Staging panen dan sortasi awal

Visual allocation 5 × 2.5 m (concept) | Fungsi disebut BP; ukuran usulan

Realistic finish study based on Blender concept; small image details may vary.

FUNCTION

Menahan sementara crate panen dan memilah awal sebelum dibawa ke hub. Bukan lini packing final penuh.

LAYOUT

Meja stainless 1,8 × 0,7 m di tepi, dua posisi trolley dan crate berlabel batch; buah utuh saja.

EQUIPMENT

Meja/crate bersih, timbangan bila diperlukan, trolley, penanda batch.

CONNECTIONS

G04 → G02 → S16 penerimaan panen → S17 QC.

SERVICES

Outlet timbangan pada sisi kering; pencatatan batch portable.

FINISHES

Permukaan halus yang dapat dibersihkan, matte light-gray; clean bays dipisahkan dari returns. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

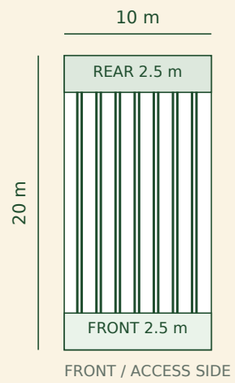
DESIGN REVIEW

Staging singkat, bukan packing final. Jangan menambah stok sampai lintasan tertutup.

SOURCE / STATUS

Bab V [0884], Gambar denah blok [0879].

DIMENSIONED CONCEPT



Same 10 × 20 m module for G01–G12.

14 rows × 30 planting anchors.

Front / crop / rear: 2.5 / 15 / 2.5 m.

150 m² gross crop band; 140 m² net unresolved.

Overlays and vignettes are not additive areas.

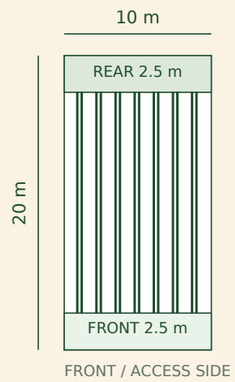
G03 / Modul produksi

Transit operator dan alat harian

Visual allocation 2.5 × 2.5 m (concept) | Usulan fungsi

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



FRONT / ACCESS SIDE

Same 10 × 20 m module for G01–G12.

14 rows × 30 planting anchors.

Front / crop / rear: 2.5 / 15 / 2.5 m.

150 m² gross crop band; 140 m² net unresolved.

Overlays and vignettes are not additive areas.

FUNCTION

Ruang lewat depan serta penyimpanan alat harian dalam kabinet. Tidak menyimpan stok pestisida di sini.

LAYOUT

Kabinet kedalaman 0,45 m dan peg rail tertutup pada tepi; satu parking trolley; alat kontak buah terpisah.

EQUIPMENT

Kabinet alat bersih, gunting panen, sarung tangan, titik parkir trolley.

CONNECTIONS

G01 ↔ G05; alat dicuci/dirawat di S09/S24.

SERVICES

Task light; charging portable di lokasi kering.

FINISHES

Matte light-gray dry floor; threshold rata; furniture mudah dibersihkan. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

Kabinet bukan store pestisida. Tidak memakai vestibule sebagai gudang.

SOURCE / STATUS

Bab V: staging dan jalur kerja; detail G03 = usulan.

G04 / Modul produksi

Zona budidaya / 7 pasang baris DFT

Visual allocation 10 × 15 m (concept) | 14 baris/420 titik di BP; pembagian panjang usulan

Realistic finish study based on Blender concept; small image details may vary.

FUNCTION

Menumbuhkan 420 tanaman, satu tanaman satu buah. 14 baris × 30 titik.

LAYOUT

14 baris / 7 pasang, 30 titik per baris, pitch 0,50 m; DFT 3 inci pada 0,8 m; training dan sling sampai trellis 2,5 m.

EQUIPMENT

Pipa DFT 3 inci pada support 0,8 m, netpot, trellis 2,5 m, sling buah.

CONNECTIONS

G06/G10 memasok larutan; G05 untuk pemeliharaan; G02 untuk panen.

SERVICES

Supply dan return ke G06; pruning, pollination dan fruit support mengikuti agronomis.

FINISHES

Permukaan existing roof dilindungi sesuai gedung; jalur abu muda dan planter/DFT support pada skid terpisah. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

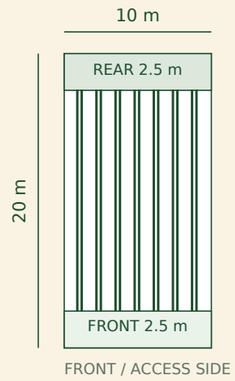
DESIGN REVIEW

420 titik dimodelkan persis. Pair envelope 0,50 m masih konsep; tajuk dan volume akar perlu trial. 150 m² bruto bukan 140 m² net.

SOURCE / STATUS

Bab V [0881], [0888], Tabel 5.2; BOM [1515].

DIMENSIONED CONCEPT



Same 10 × 20 m module for G01–G12.

14 rows × 30 planting anchors.

Front / crop / rear: 2.5 / 15 / 2.5 m.

150 m² gross crop band; 140 m² net unresolved.

Overlays and vignettes are not additive areas.

G05 / Modul produksi

Jalur kerja dan akses tepi

Visual allocation 10 × 15 m (concept) | 80 cm di BP; jalur tepi usulan

Realistic finish study based on Blender concept; small image details may vary.

FUNCTION

Penyerbukan, pruning, inspeksi, perawatan, pengambilan buah dan akses utilitas.

LAYOUT

Enam jalur antar-pasang 0,80 m, dua tepi 0,85 m; marking tipis; semua selang melintas melalui bridge/tepi.

EQUIPMENT

Jalur tanpa stok/barang; permukaan antiselip; trolley sesuai lebar efektif.

CONNECTIONS

Zona depan ↔ G04 sepanjang baris ↔ utilitas belakang.

SERVICES

Lampu kerja dan permukaan antiselip; trolley ramping yang benar-benar muat setelah tajuk tumbuh.

FINISHES

Permukaan existing roof dilindungi sesuai gedung; jalur abu muda dan planter/DFT support pada skid terpisah. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

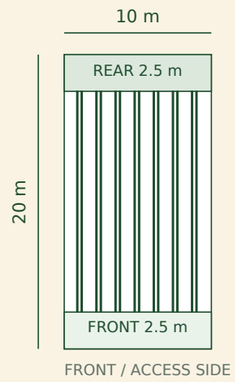
DESIGN REVIEW

Jangan memakai 0,80 m sebagai klaim aksesibilitas atau egress yang disetujui. Lebar bersih diukur setelah tajuk dan equipment.

SOURCE / STATUS

Bab V [0882]; 6 jalur berasal dari 7 pasang. Dua jalur tepi = usulan.

DIMENSIONED CONCEPT



Same 10 × 20 m module for G01–G12.

14 rows × 30 planting anchors.

Front / crop / rear: 2.5 / 15 / 2.5 m.

150 m² gross crop band; 140 m² net unresolved.

Overlays and vignettes are not additive areas.

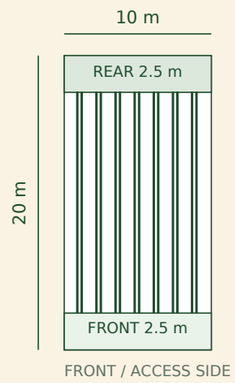
G06 / Modul produksi

Zona basah nutrisi / sirkulasi

Visual allocation 5 × 2.5 m (concept) | Fungsi BP; pembagian sistem perlu klarifikasi

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



Same 10 × 20 m module for G01–G12.

14 rows × 30 planting anchors.

Front / crop / rear: 2.5 / 15 / 2.5 m.

150 m² gross crop band; 140 m² net unresolved.

Overlays and vignettes are not additive areas.

FUNCTION

Mengontrol larutan kerja, level, aliran dan kembalian DFT. Jangan menyamakan dua stock A/B dengan reservoir kerja.

LAYOUT

Dua reservoir nominal 500 L, bund rendah, dua pump skids, manifold dan dua stock concentrates kecil yang terpisah.

EQUIPMENT

BOM: dua bak 500 L, dua pompa 25 W. Usulan: reservoir kerja, stock A/B terpisah, valve, filter, secondary containment.

CONNECTIONS

S06 air → S07 stok / persiapan → G06 reservoir kerja → G10 DFT → return.

SERVICES

Isolasi loop, drain terkendali, sampling bypass, pH/EC pada larutan; stock A/B tidak dicampur sebelum dosing.

FINISHES

Lantai antiselip warna abu muda dengan sambungan bersih dan drain terkendali; detail kemiringan belum final. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

Peran dua tank BP belum final. Render memilih reservoir kerja + stok kecil tambahan sebagai usulan; bukan BOM final. Flow/head dan beban slab belum dihitung.

SOURCE / STATUS

Bab V [0883], [0968-0969], BOM; prinsip stok terpisah: Penn State E2.

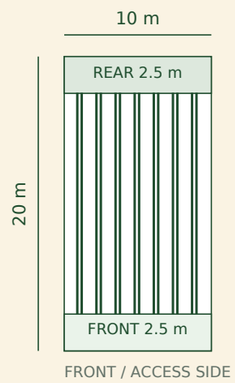
G07 / Modul produksi

Zona mekanikal dan service iklim

Visual allocation 3 × 2.5 m (concept) | Peralatan BP; luas usulan

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



Same 10 × 20 m module for G01–G12.

14 rows × 30 planting anchors.

Front / crop / rear: 2.5 / 15 / 2.5 m.

150 m² gross crop band; 140 m² net unresolved.

Overlays and vignettes are not additive areas.

FUNCTION

Menempatkan peralatan yang membutuhkan akses perawatan, tanpa menutup jalur kerja.

LAYOUT

Dehumidifier dengan grille terbuka, filter dapat dilepas; condensate ke drain terkendali; working face ditandai.

EQUIPMENT

Dehumidifier satu unit; manifolds/peralatan mekanis; jalur kondensat. Blower berada pada dinding.

CONNECTIONS

G12 iklim; kondensat ke sistem drain yang sesuai; akses service dari jalur belakang.

SERVICES

Koneksi power kering; condensate; dua blower tetap pada dinding GH.

FINISHES

Matte light-gray dry floor; threshold rata; furniture mudah dibersihkan. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

Footprint alat dan jarak servis harus memakai shop drawing vendor; kapasitas tidak ditentukan dari model visual.

SOURCE / STATUS

Bab V [0883], [0970-0975]; CDC E1 untuk genset.

G08 / Modul produksi

Panel listrik / kontrol kering

Visual allocation 2 × 2.5 m (concept) | Panel BP; pemisahan kering usulan

Realistic finish study based on Blender concept; small image details may vary.

FUNCTION

Kontrol dan proteksi rangkaian; akses teknisi pada bagian kering. Tidak otomatis ruangan tertutup.

LAYOUT

Kabinet modul 0,30 × 0,30 m pada dinding kering; raceway di atas splash zone; diagram koneksi terpisah dari wet work.

EQUIPMENT

Panel BOM 30 × 30 cm, controller ESP32, relay, network/power protection sesuai desain listrik.

CONNECTIONS

S10 panel site/IT → G08 → pompa, blower, sensor; data ke dashboard.

SERVICES

Controller ESP32 dan relay dari BOM; isolasi, grounding dan enclosure oleh engineer.

FINISHES

Matte light-gray dry floor; threshold rata; furniture mudah dibersihkan. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

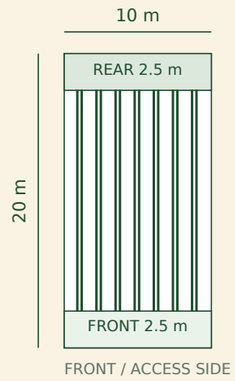
DESIGN REVIEW

Model menunjukkan casing dan jalur, bukan wiring diagram. Panel besar site ada di S10.

SOURCE / STATUS

Bab V [0883], BOM [1515], CAPEX_Detail B10/B12.

DIMENSIONED CONCEPT



FRONT / ACCESS SIDE

Same 10 × 20 m module for G01–G12.

14 rows × 30 planting anchors.

Front / crop / rear: 2.5 / 15 / 2.5 m.

150 m² gross crop band; 140 m² net unresolved.

Overlays and vignettes are not additive areas.

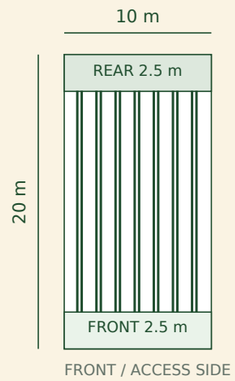
G09 / Modul produksi

Selubung, struktur, atap dan trellis

Visual allocation 10 × 20 m (concept) | Dimensi dan komponen BP

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



Same 10 × 20 m module for G01–G12.

14 rows × 30 planting anchors.

Front / crop / rear: 2.5 / 15 / 2.5 m.

150 m² gross crop band; 140 m² net unresolved.

Overlays and vignettes are not additive areas.

FUNCTION

Mendukung cover, peralatan dan tanaman, serta membatasi masuknya hama/cuaca.

LAYOUT

Portal galvanis berulang, eave 3 m / ridge 4,5 m; UV-film konsep; insect-mesh sides; trellis terpisah dari cover.

EQUIPMENT

Rangka galvanis, UV film atau polycarbonate (opsi belum diputuskan), insect net 40 mesh, trellis/sling.

CONNECTIONS

Terhubung load path ke struktur gedung, roof waterproofing dan drain eksternal.

SERVICES

Roof water ke tepi/drain gedung; screen openings; anchors dan bracing mengikuti engineer.

FINISHES

Permukaan existing roof dilindungi sesuai gedung; jalur abu muda dan planter/DFT support pada skid terpisah. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

Film dan mesh visual bukan detail sambungan. 208,8 m² bidang atap sebelum overlap; struktur rooftop belum disetujui.

SOURCE / STATUS

Bab V [0878], [0888], BOM; shell area reconciliation pada catatan dimensi.

G10 / Modul produksi

Jalur supply, return dan drainase

Visual allocation 5 × 2.5 m (concept) | Resirkulasi BP; detail jaringan usulan

Realistic finish study based on Blender concept; small image details may vary.

FUNCTION

Mendistribusikan dan mengembalikan larutan kerja; air hujan, limbah nutrisi dan kondensat dipetakan tersendiri.

LAYOUT

Supply manifold pada satu sisi, return terpisah, valve per loop dan sampling bypass pada wet zone; clean-out dapat dijangkau.

EQUIPMENT

Supply/return manifold, valve per loop, filter, titik inspeksi, saluran dan akses bersih.

CONNECTIONS

G06 ↔14 baris; kondensat G07; drain bangunan S27.

SERVICES

DFT circulation; garis biru = konsep supply, abu = return, orange = dosing.

FINISHES

Lantai antiselip warna abu muda dengan sambungan bersih dan drain terkendali; detail kemiringan belum final. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

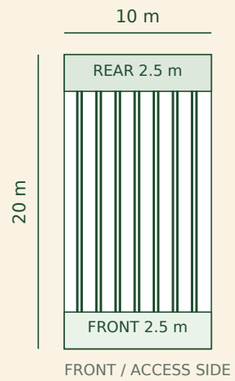
DESIGN REVIEW

Warna adalah wayfinding konsep, bukan standar plumbing. Diameter, slope, head dan disinfection belum dirancang.

SOURCE / STATUS

Bab V [0968-0969], DFT risk tables, BOM PVC/fitting; isolation = usulan.

DIMENSIONED CONCEPT



FRONT / ACCESS SIDE

Same 10 × 20 m module for G01–G12.

14 rows × 30 planting anchors.

Front / crop / rear: 2.5 / 15 / 2.5 m.

150 m² gross crop band; 140 m² net unresolved.

Overlays and vignettes are not additive areas.

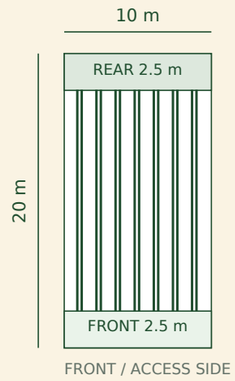
G11 / Modul produksi

Sensor dan titik monitoring

Visual allocation 4 × 3 m (concept) | Qty BOM di BP; posisi rinci usulan

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



FRONT / ACCESS SIDE

Same 10 × 20 m module for G01–G12.

14 rows × 30 planting anchors.

Front / crop / rear: 2.5 / 15 / 2.5 m.

150 m² gross crop band; 140 m² net unresolved.

Overlays and vignettes are not additive areas.

FUNCTION

Membandingkan kondisi depan-tengah-belakang dan kualitas larutan sirkulasi.

LAYOUT

Tiga ambient T/RH positions depan-tengah-belakang; shield pada bracket. Detail pH/EC dalam bypass larutan dan flow pada pipa.

EQUIPMENT

3 suhu/RH, 2 pH, 2 EC/TDS, 1 flow sensor pada BOM. Posisi dan jumlah loop belum diputuskan.

CONNECTIONS

Ambient ke G08; pH/EC/flow pada G06/G10; data ke S10.

SERVICES

Low voltage cable tray ke G08; servis dan calibration ports terlihat.

FINISHES

Matte light-gray dry floor; threshold rata; furniture mudah dibersihkan. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

Vignette menunjukkan hardware tanpa angka live. Detail loop kedua tetap memerlukan hydraulic schematic.

SOURCE / STATUS

Bab V [0886], [0963-0967], BOM [1515].

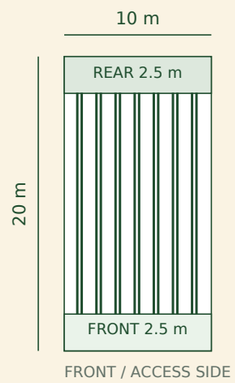
G12 / Modul produksi

Ventilasi dan kontrol iklim

Visual allocation 3 × 2.5 m (concept) | Dua blower/dehumidifier BP

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



Same 10 × 20 m module for G01–G12.

14 rows × 30 planting anchors.

Front / crop / rear: 2.5 / 15 / 2.5 m.

150 m² gross crop band; 140 m² net unresolved.

Overlays and vignettes are not additive areas.

FUNCTION

Sirkulasi udara dan pengendalian kelembapan mengikuti target agronomi.

LAYOUT

Dua axial blowers dinding, insect-barrier intake dan dehumidifier; arah aliran ditinjau tanpa mengklaim kinerja.

EQUIPMENT

Dua blower di dinding kiri/kanan; dehumidifier; inlet/outlet dan insect barrier sesuai desain.

CONNECTIONS

G07 mekanikal, G09 envelope, G11 sensor.

SERVICES

Power, condensate dan akses filter; control dari G08.

FINISHES

Permukaan existing roof dilindungi sesuai gedung; jalur abu muda dan planter/DFT support pada skid terpisah. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

GH tetap ruang budidaya dengan screen/film, bukan ruang AC tertutup. Target suhu/RH memerlukan sizing panas dan kelembapan.

SOURCE / STATUS

Bab V [0885], [0972-0975]; airflow/thermal sizing belum ada.

S01 / Bibit dan nursery

Penerimaan input / goods receiving

Visual allocation 4 × 4 m (concept) | Usulan ruang bersama

Realistic finish study based on Blender concept; small image details may vary.

FUNCTION

Menerima benih, nutrisi, substrate dan consumables; cek batch/kondisi sebelum masuk gudang.

LAYOUT

Landing crate/pallet di sisi pintu; meja inspection 1,6 m; rak hold singkat; alur langsung ke S02/S07/S08.

EQUIPMENT

Meja inspeksi, pallet/rak sementara, log penerimaan.

CONNECTIONS

S26 akses barang → S01 → S02/S07/S08.

SERVICES

Lighting inspeksi; scanner/log portable; pintu servis.

FINISHES

Matte light-gray dry floor; threshold rata; furniture mudah dibersihkan. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

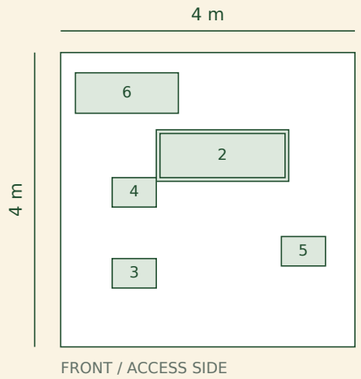
DESIGN REVIEW

16 m² studi ruang. Tidak menaruh input kotor pada penerimaan buah S16.

SOURCE / STATUS

Supply network Bab V; ruang receiving = usulan.

DIMENSIONED CONCEPT



1. Inspection top · 1.8 × 0.7 m
2. Inspection shelf · 1.7 × 0.6 m
3. Receiving crate base · 0.6 × 0.4 m
4. Receiving crate base.001 · 0.6 × 0.4 m
5. Receiving crate base.002 · 0.6 × 0.4 m
6. Hold Rack shelf · 1.4 × 0.55 m

S02 / Bibit dan nursery

Gudang benih dan input kering

Visual allocation 3 × 3 m (concept) | Usulan dari kebutuhan stok

Realistic finish study based on Blender concept; small image details may vary.

FUNCTION

Menjaga benih, netpot dan input sensitif tetap kering dan teridentifikasi.

LAYOUT

Rak tertutup kedalaman 0,45 m pada back wall; benih dalam bin tertutup, lot dan expiry terpisah dari nutrisi.

EQUIPMENT

Rak tertutup, lot/expiry tracking, penyimpanan sesuai supplier.

CONNECTIONS

S01 → S02 → S03; consumables ke S08.

SERVICES

Monitoring kondisi simpan sesuai supplier; task light.

FINISHES

Matte light-gray dry floor; threshold rata; furniture mudah dibersihkan. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

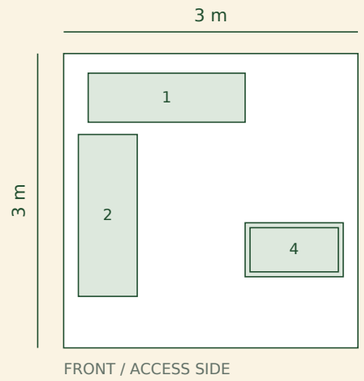
DESIGN REVIEW

9 m² konsep, bukan volume stok terhitung. Suhu simpan benih belum ditetapkan.

SOURCE / STATUS

Panduan Bab V inventori; supply network benih; fungsi ruang = usulan.

DIMENSIONED CONCEPT



1. Storage shelf · 1.6 × 0.5 m
2. Side Storage shelf · 0.6 × 1.65 m
3. Inventory Desk top · 1 × 0.55 m
4. Inventory Desk shelf · 0.9 × 0.45 m

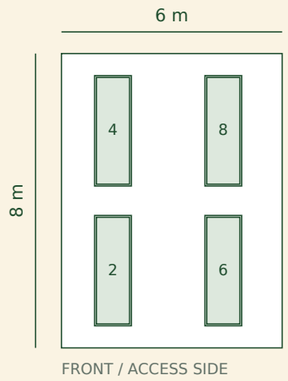
S03 / Bibit dan nursery

Persemaian / nursery

Visual allocation 6 × 8 m (concept) | Usulan ruang produksi bersama

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



1. Nursery Bench top · 1 × 3 m
2. Nursery Bench shelf · 0.9 × 2.9 m
3. Nursery Bench top.001 · 1 × 3 m
4. Nursery Bench shelf.001 · 0.9 × 2.9 m
5. Nursery Bench top.002 · 1 × 3 m
6. Nursery Bench shelf.002 · 0.9 × 2.9 m
7. Nursery Bench top.003 · 1 × 3 m
8. Nursery Bench shelf.003 · 0.9 × 2.9 m

FUNCTION

Germinasi dan pembesaran bibit sebelum transplantasi. Tidak otomatis memakai lantai 200 m² modul panen.

LAYOUT

Empat benches 1 × 3 m, tray plug dan rockwool; cross aisle depan; mesh enclosure dan transplant work surface.

EQUIPMENT

Tray/rockwool, bench, supply air/air bersih, alat monitoring.

CONNECTIONS

S02 benih → S03 → S04 → G04.

SERVICES

Air bersih dengan supply terkendali; ambient monitor; tray cleaning di S24.

FINISHES

Permukaan existing roof dilindungi sesuai gedung; jalur abu muda dan planter/DFT support pada skid terpisah. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

48 m² konsep. Tray count, seedling age, cadangan dan overlap batch belum dihitung; tidak menyatakan cukup untuk 17/18 GH.

SOURCE / STATUS

Benih/substrate dan proses tanam pada Bab V; nursery terpisah belum dirinci.

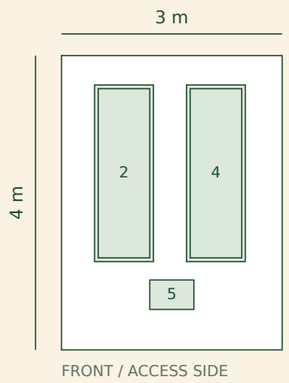
S04 / Bibit dan nursery

Area bibit siap transplantasi

Visual allocation 3 × 4 m (concept) | Usulan zona; dapat menyatu dengan nursery

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



1. Ready Batch top · 0.8 × 2.4 m
2. Ready Batch shelf · 0.7 × 2.3 m
3. Ready Batch top.001 · 0.8 × 2.4 m
4. Ready Batch shelf.001 · 0.7 × 2.3 m
5. Transplant Trolley crate base · 0.6 × 0.4 m

FUNCTION

Hold bibit yang sudah siap dan cek identitas batch untuk GH tujuan.

LAYOUT

Dua benches pendek, satu trolley bibit dan posisi batch release; rute paling singkat ke pintu GH.

EQUIPMENT

Bench terlabel, trolley bersih, checklist kondisi bibit.

CONNECTIONS

S03 → S04 → G01/G04.

SERVICES

Supply air/air bersih sesuai nursery; portable checklist.

FINISHES

Permukaan existing roof dilindungi sesuai gedung; jalur abu muda dan planter/DFT support pada skid terpisah. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

12 m² konsep untuk hold singkat; jangan menjadi persemaian kedua tanpa menghitung kapasitas.

SOURCE / STATUS

Usulan alur operasi dari tahapan budidaya.

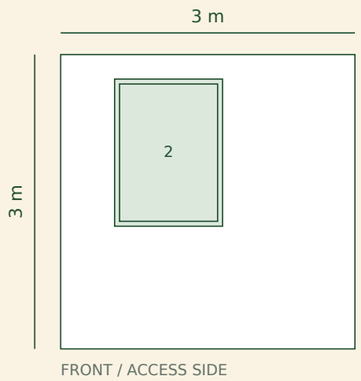
S05 / Bibit dan nursery

Observasi tanaman / isolasi IPM

Visual allocation 3 × 3 m (concept) | Usulan zona terisolasi

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



1. IPM Observation top · 1.1 × 1.5 m
2. IPM Observation shelf · 1 × 1.4 m

FUNCTION

Menahan bibit/tanaman suspek untuk observasi tanpa memasukkan kembali ke alur sehat.

LAYOUT

Single inspection bench, specimen hold dan tools khusus; akses terkendali; pemisahan tanaman sakit jelas.

EQUIPMENT

Bench isolasi, alat khusus, log observasi dan akses terbatas.

CONNECTIONS

S03/G04 → S05 → keputusan agronomi/S25.

SERVICES

Ventilasi yang tidak diarahkan ke nursery sehat; catatan IPM.

FINISHES

Permukaan existing roof dilindungi sesuai gedung; jalur abu muda dan planter/DFT support pada skid terpisah. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

9 m² konsep. Tidak mengklaim containment biologis; protokol pest/agronomy menentukan isolasi sebenarnya.

SOURCE / STATUS

Risiko penyakit DFT pada Bab V; konfigurasi isolasi = usulan.

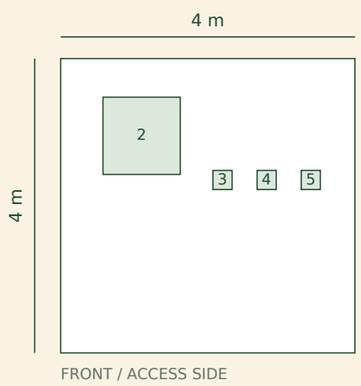
S06 / Utilitas dan pemeliharaan

Air baku / buffer / treatment

Visual allocation 4 × 4 m (concept) | Usulan ruang/area utilitas bersama

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



1. Water Buffer vessel · 1 × 1 m
2. Water Buffer base · 1.05 × 1.05 m
3. Filter Housing · 0.26 × 0.26 m
4. Filter Housing.001 · 0.26 × 0.26 m
5. Filter Housing.002 · 0.26 × 0.26 m

FUNCTION

Menyediakan air dengan mutu dan kontinuitas yang cocok untuk hidroponik.

LAYOUT

Buffer vessel, filter housings, meter dan sampling point pada skid; aisle servis pada sisi valves. Akses servis dari depan; model revisi tidak memakai pintu belakang.

EQUIPMENT

Buffer tank, water meter, treatment yang dipilih setelah uji air.

CONNECTIONS

Sumber air → S06 → S07/G06; roof runoff tidak otomatis dipakai.

SERVICES

Incoming water; quality test sebelum treatment; drain terkendali.

FINISHES

Lantai antiselip warna abu muda dengan sambungan bersih dan drain terkendali; detail kemiringan belum final. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

16 m² visual allocation. Volume tank di render placeholder equipment envelope; uji air dan struktur menentukan kapasitas.

SOURCE / STATUS

Sumber air Panduan Bab V; sizing dan treatment = usulan.

S07 / Utilitas dan pemeliharaan

Preparation nutrisi dan store kimia

Visual allocation 3 × 4 m (concept) | Usulan pemisahan fungsi bersama

Realistic finish study based on Blender concept; small image details may vary.

FUNCTION

Menyimpan dan menyiapkan nutrisi/produk budidaya secara terkontrol; stock A/B terpisah.

LAYOUT

Dua covered stock stations A/B terpisah, bench penakaran, closed cabinets dan secondary containment. Furniture/skid ditarik dari pendekatan pintu belakang pada revisi model.

EQUIPMENT

Rak dan bund/containment sesuai bahan, alat ukur, label, area mix, fasilitas penanganan sesuai SOP.

CONNECTIONS

S01 input → S07 → G06; limbah diarahkan sesuai prosedur.

SERVICES

Water point dan penanganan bahan sesuai SDS/SOP; dosing supply ke G06.

FINISHES

Lantai antiselip warna abu muda dengan sambungan bersih dan drain terkendali; detail kemiringan belum final. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

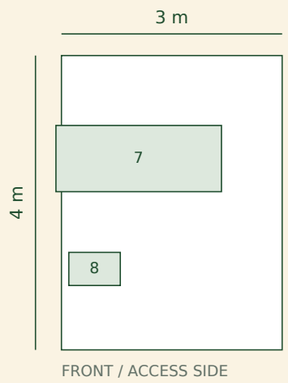
DESIGN REVIEW

12 m² konsep. Kompatibilitas, ventilasi dan fasilitas penanganan ditetapkan bahan aktual; tidak mencampur chemicals di rak buah.

SOURCE / STATUS

AB Mix Bab V dan BOM; prinsip A/B: Penn State E2.

DIMENSIONED CONCEPT



1. Mixing Bench top · 2 × 0.65 m
2. Mixing Bench shelf · 1.9 × 0.55 m
3. Stock A vessel.001 · 0.36 × 0.36 m
4. Stock A base.001 · 0.378 × 0.378 m
5. Stock B vessel.001 · 0.36 × 0.36 m
6. Stock B base.001 · 0.378 × 0.378 m
7. Mixing Bund · 2.25 × 0.9 m
8. Chemical Cabinet · 0.7 × 0.45 m

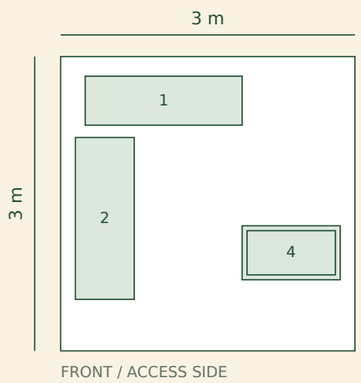
S08 / Utilitas dan pemeliharaan

Gudang alat / suku cadang

Visual allocation 3 × 3 m (concept) | Peralatan didanai; ruang usulan

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



1. Storage shelf.004 · 1.6 × 0.5 m
2. Side Storage shelf.004 · 0.6 × 1.65 m
3. Inventory Desk top.001 · 1 × 0.55 m
4. Inventory Desk shelf.001 · 0.9 × 0.45 m

FUNCTION

Menjaga spare pompa, sensor, fittings dan tools agar penggantian cepat.

LAYOUT

Rak parts bins, spare pumps, fittings dan probes dalam tempat pelindung; check-out bench kecil.

EQUIPMENT

Rak item kritis, parts bin, alat cadangan dan log.

CONNECTIONS

S01 → S08 → S09/G03/G08.

SERVICES

Light dan inventory log; no wet washing.

FINISHES

Matte light-gray dry floor; threshold rata; furniture mudah dibersihkan. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

9 m² konsep. Jaga spares kritis berbeda dari tools kotor.

SOURCE / STATUS

CAPEX_Detail B38; risk supply network; gudang = usulan.

S09 / Utilitas dan pemeliharaan

Workshop / maintenance bench

Visual allocation 4 × 4 m (concept) | Usulan area bersama

Realistic finish study based on Blender concept; small image details may vary.

FUNCTION

Pemeriksaan pompa/sensor, kalibrasi dan perbaikan ringan di luar zona buah.

LAYOUT

Workbench 2,2 m, pegboard, servis pump bench dan calibration station; alat kotor masuk melalui S24 bila relevan. Koreksi model: rak parts dipindahkan ke sisi kiri agar pendekatan pintu belakang bebas.

EQUIPMENT

Workbench, alat servis, calibration reference sesuai sensor, tray parts.

CONNECTIONS

S08 spare ↔ S09 ↔ G06/G08; tools kotor ke S24.

SERVICES

Power bench, task light, outlet proteksi oleh engineer.

FINISHES

Matte light-gray dry floor; threshold rata; furniture mudah dibersihkan. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

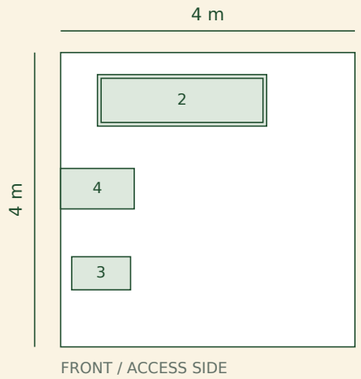
DESIGN REVIEW

16 m² konsep; workshop ringan. Tidak memberi izin welding/hot work pada rooftop.

SOURCE / STATUS

Maintenance supplier Bab V; area kerja = usulan.

DIMENSIONED CONCEPT



1. Service Bench top · 2.3 × 0.7 m
2. Service Bench shelf · 2.2 × 0.6 m
3. Tool Cabinet · 0.8 × 0.45 m
4. Parts shelf · 1 × 0.55 m

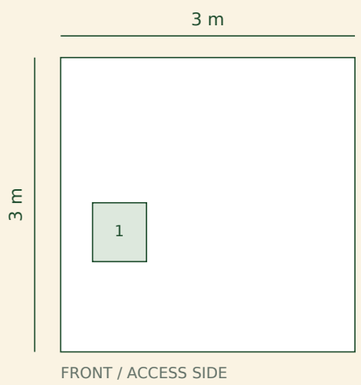
S10 / Utilitas dan pemeliharaan

Panel site / jaringan / UPS

Visual allocation 3 × 3 m (concept) | Infrastruktur IT didanai; ruang usulan

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



1. Network Rack · 0.55 × 0.6 m

FUNCTION

Distribusi listrik site dan koneksi data antarmodul; lokasi NAS pusat dapat berbeda.

LAYOUT

Site distribution panel, network rack dan UPS; front service zone terpisah dari kotak stok. Rack jaringan dipindahkan ke kiri depan agar pintu belakang bebas.

EQUIPMENT

Panel site, network switch/gateway, UPS, local data access; NAS pusat pada CAPEX.

CONNECTIONS

Listrik/data → S10 → G08/G11; akses admin S15.

SERVICES

Data gateway/NAS sesuai lokasi aktual; circuit segregation; heat rejection vendor.

FINISHES

Matte light-gray dry floor; threshold rata; furniture mudah dibersihkan. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

9 m² konsep. NAS pusat tidak diduplikasi di tiap GH; UPS bukan backup semua climate load.

SOURCE / STATUS

CAPEX_Detail B39; detail distribusi listrik = usulan.

S11 / Utilitas dan pemeliharaan

Genset / power backup eksternal

Visual allocation 3 × 4 m (concept) | Peralatan BP; relokasi operasional diperlukan

Realistic finish study based on Blender concept; small image details may vary.

FUNCTION

Menopang beban kritis ketika listrik padam. Genset bensin tidak beroperasi di GH atau ruang tertutup.

LAYOUT

Genset kecil pada plinth, open-sided shelter dan access guard; tidak enclosed; air intake/exhaust tidak menuju bukaan.

EQUIPMENT

BOM: genset 1.200 W. Desain backup/load transfer sesuai electrical engineer.

CONNECTIONS

S11 → distribusi listrik → critical pump/control circuits.

SERVICES

Transfer/load design engineer; runoff tidak masuk drainage buah.

FINISHES

Matte light-gray dry floor; threshold rata; furniture mudah dibersihkan. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

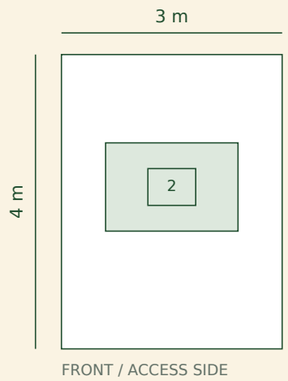
DESIGN REVIEW

12 m² adalah vignette, bukan batas exhaust clearance. Lokasi operasi ≥6,1 m dari bukaan sesuai CDC; tidak memuat depot BBM.

SOURCE / STATUS

Bab V [0961-0962], BOM; CDC/NIOSH E1. Ini batas prinsip, bukan gambar instalasi.

DIMENSIONED CONCEPT



1. Generator Plinth · 1.8 × 1.2 m
2. Small Generator · 0.65 × 0.5 m

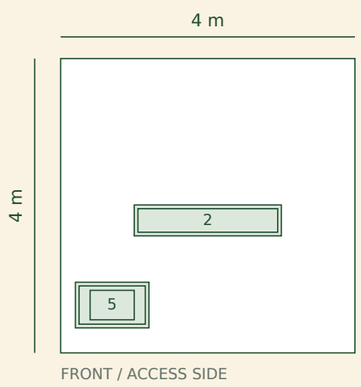
S12 / Staf dan administrasi

Locker / change / higiene staf

Visual allocation 4 × 4 m (concept) | Usulan fasilitas bersama

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



1. Changing Bench top · 2 × 0.42 m
2. Changing Bench shelf · 1.9 × 0.32 m
3. Hygiene top · 1 × 0.62 m
4. Hygiene shelf · 0.9 × 0.52 m
5. Hygiene basin · 0.6 × 0.4 m

FUNCTION

Membuat transisi pakaian/alat kaki dan higiene sebelum masuk budidaya atau packing.

LAYOUT

Locker baris belakang, changing bench, handwash dan APD; transisi personal belongings ke uniform bersih. Furniture/skid ditarik dari pendekatan pintu belakang pada revisi model.

EQUIPMENT

Locker, changing bench, higiene tangan sesuai SOP, tempat APD.

CONNECTIONS

Staff entry → S12 → G01 atau S19 melalui jalur yang sesuai.

SERVICES

Supply air bersih, wastewater dan ventilasi gedung.

FINISHES

Lantai antiselip warna abu muda dengan sambungan bersih dan drain terkendali; detail kemiringan belum final. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

16 m² konsep. Jumlah locker berdasarkan staf peak aktual, bukan semua headcount perusahaan.

SOURCE / STATUS

SDM/SOP hygiene pada BP; room program = usulan.

S13 / Staf dan administrasi

Toilet / sanitasi staf

Visual allocation 3 × 4 m (concept) | Usulan; dapat menggunakan fasilitas gedung

Realistic finish study based on Blender concept; small image details may vary.

FUNCTION

Menyediakan kebutuhan staf di luar ruang budidaya dan packing.

LAYOUT

Dua cubicle konsep dan handwash lobby; doorway dipisah dari zona buah; finishes yang dapat dibersihkan. Dua panel privasi cubicle; akses masuk dari depan, tanpa pintu servis belakang.

EQUIPMENT

Fasilitas sanitasi dan higiene tangan gedung/site.

CONNECTIONS

S12/S14 → S13 → transisi higiene sebelum kembali kerja.

SERVICES

Sanitary plumbing dan exhaust gedung.

FINISHES

Lantai antiselip warna abu muda dengan sambungan bersih dan drain terkendali; detail kemiringan belum final. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

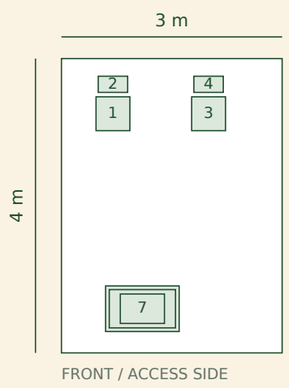
DESIGN REVIEW

12 m² studi. Accessible toilet, jumlah fixture dan jalur akses final mengikuti gedung/regulasi; ukuran cubicle visual belum approval.

SOURCE / STATUS

Fasilitas gedung belum dirinci pada sumber; usulan program.

DIMENSIONED CONCEPT



1. WC · 0.46 × 0.46 m
2. WC Cistern · 0.4 × 0.22 m
3. WC.001 · 0.46 × 0.46 m
4. WC Cistern.001 · 0.4 × 0.22 m
5. Handwash top · 1 × 0.62 m
6. Handwash shelf · 0.9 × 0.52 m
7. Handwash basin · 0.6 × 0.4 m

S14 / Staf dan administrasi

Istirahat / briefing / P3K

Visual allocation 4 × 5 m (concept) | Usulan ruang bersama

Realistic finish study based on Blender concept; small image details may vary.

FUNCTION

Ruang staf untuk break, briefing dan perlengkapan pertolongan pertama.

LAYOUT

Meja briefing, bench/seat, SOP board dan first-aid cabinet; tidak menjadi pantry di packing bersih.

EQUIPMENT

Meja/seat, board SOP, perlengkapan P3K sesuai site.

CONNECTIONS

S12, S15 dan akses keluar site; di luar crop/pack.

SERVICES

Light, ventilasi dan water point bila gedung menyediakan.

FINISHES

Matte light-gray dry floor; threshold rata; furniture mudah dibersihkan. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

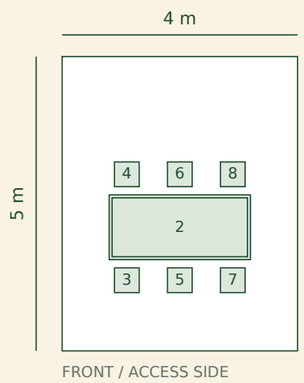
DESIGN REVIEW

20 m² konsep. Isi P3K dan kapasitas kursi mengikuti SOP dan staf shift.

SOURCE / STATUS

SDM/briefing pada BP; ruang = usulan.

DIMENSIONED CONCEPT



1. Briefing Table top · 2.4 × 1.1 m
2. Briefing Table shelf · 2.3 × 1 m
3. Briefing Seat seat · 0.42 × 0.42 m
4. Briefing Seat seat.001 · 0.42 × 0.42 m
5. Briefing Seat seat.002 · 0.42 × 0.42 m
6. Briefing Seat seat.003 · 0.42 × 0.42 m
7. Briefing Seat seat.004 · 0.42 × 0.42 m
8. Briefing Seat seat.005 · 0.42 × 0.42 m

S15 / Staf dan administrasi

Kantor kecil / admin site

Visual allocation 3 × 3 m (concept) | Disebut eksplisit workbook

Realistic finish study based on Blender concept; small image details may vary.

FUNCTION

Catatan operasi, produksi, order, stok dan koordinasi dua lokasi.

LAYOUT

Dua workstation compact, printer cabinet dan wall planning board; sightline ke akses site. Furniture/skid ditarik dari pendekatan pintu belakang pada revisi model.

EQUIPMENT

CAPEX: meja/kursi, AC,2 laptop, printer; akses dashboard.

CONNECTIONS

S10 data; S17 QC; S23 dispatch; S14 briefing.

SERVICES

Power/data/AC dari CAPEX; monitor berisi layout dan chart simulasi.

FINISHES

Matte light-gray dry floor; threshold rata; furniture mudah dibersihkan. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

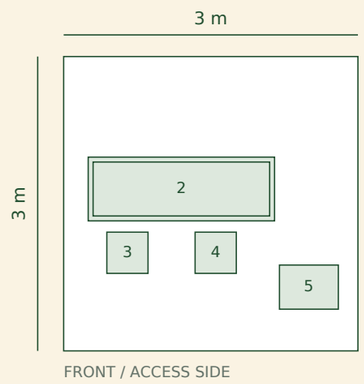
DESIGN REVIEW

9 m² dari workbook; 3 × 3 m adalah bentuk konsep. Tidak tampilkan data akun atau dashboard live.

SOURCE / STATUS

CAPEX_Detail B37/N37: office build-out tiap lokasi 9 m².

DIMENSIONED CONCEPT



1. Desk top · 1.9 × 0.65 m
2. Desk shelf · 1.8 × 0.55 m
3. Office Chair seat · 0.42 × 0.42 m
4. Office Chair seat.001 · 0.42 × 0.42 m
5. Printer Cabinet · 0.6 × 0.45 m

S16 / Pascapanen dan hub

Penerimaan buah panen di hub

Visual allocation 4 × 5 m (concept) | Usulan zona dari alur BP

Realistic finish study based on Blender concept; small image details may vary.

FUNCTION

Menerima crate per GH/batch dan menjaga identitas buah sebelum QC.

LAYOUT

Clean crate inbound, platform scale dan batch desk; waiting positions sebelum QC; terpisah dari dirty crate return.

EQUIPMENT

Crate bersih, tempat timbang, meja data, trolley.

CONNECTIONS

G02 → S16 → S17.

SERVICES

Data capture portable; lighting inspeksi; trolley route.

FINISHES

Permukaan halus yang dapat dibersihkan, matte light-gray; clean bays dipisahkan dari returns. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

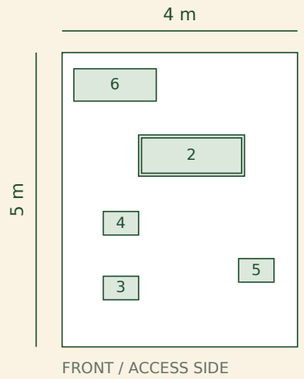
DESIGN REVIEW

20 m² konsep. Harvest peak dan lift schedule menentukan receiving buffer sebenarnya.

SOURCE / STATUS

G02 ke distribution hub Bab V [0884]; detail hub = usulan.

DIMENSIONED CONCEPT



1. Inspection top.001 · 1.8 × 0.7 m
2. Inspection shelf.001 · 1.7 × 0.6 m
3. Receiving crate base.003 · 0.6 × 0.4 m
4. Receiving crate base.004 · 0.6 × 0.4 m
5. Receiving crate base.005 · 0.6 × 0.4 m
6. Hold Rack shelf.003 · 1.4 × 0.55 m

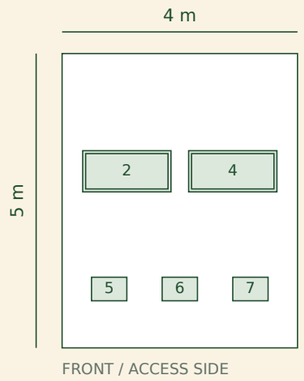
S17 / Pascapanen dan hub

QC / grading / timbang

Visual allocation 4 × 5 m (concept) | Fungsi QC BP; ruang usulan

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



1. QC Bench top · 1.5 × 0.7 m
2. QC Bench shelf · 1.4 × 0.6 m
3. QC Bench top.001 · 1.5 × 0.7 m
4. QC Bench shelf.001 · 1.4 × 0.6 m
5. Grade Bay base · 0.6 × 0.4 m
6. Grade Bay base.001 · 0.6 × 0.4 m
7. Grade Bay base.002 · 0.6 × 0.4 m

FUNCTION

Sortir mutu visual, berat dan hasil/sampling kemanisan; tetapkan release setiap batch.

LAYOUT

Dua QC benches dengan scale dan refractometer, clear inspection light; Grade A/B/C crate bays setelah keputusan release.

EQUIPMENT

Timbangan, refractometer/alat QC, meja, catatan batch.

CONNECTIONS

S16 → S17 → S18 hold atau S19 pack; Grade B/C outlet terpisah.

SERVICES

Power timbangan; data batch; sanitasi alat sesuai SOP.

FINISHES

Permukaan halus yang dapat dibersihkan, matte light-gray; clean bays dipisahkan dari returns. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

20 m² konsep. Tidak menempel angka Brix contoh sebagai hasil buah. Satu release SOP perlu menyamakan target ≥15° Bab IV dengan bagian lama.

SOURCE / STATUS

Bab IV QC/label; Bab V sortasi; alat farming CAPEX B38.

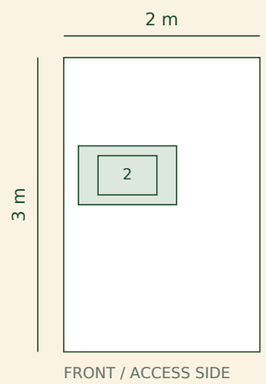
S18 / Pascapanen dan hub

Hold produk / non-release

Visual allocation 2 × 3 m (concept) | Usulan zona bertanda

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



1. Hold Only shelf · 1 × 0.6 m
2. Hold Crate base · 0.6 × 0.4 m

FUNCTION

Mencegah produk yang belum lolos QC atau menunggu keputusan tercampur produk rilis.

LAYOUT

Dedicated hold rack dengan barrier kecil dan log; dipisah dari buah released dan inventory siap dispatch. Rak hold dipersempit menjadi 1 m agar jalur kanan tetap bebas.

EQUIPMENT

Rak/crate bertanda hold, log dan kontrol akses.

CONNECTIONS

S17 → S18 → release ulang atau disposition/secondary outlet/S25.

SERVICES

Log release/quarantine; hanya staf berwenang.

FINISHES

Permukaan halus yang dapat dibersihkan, matte light-gray; clean bays dipisahkan dari returns. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

6 m² konsep, dapat berupa zona bukan kamar. Suhu hold mengikuti keputusan QC dan cold-chain.

SOURCE / STATUS

Usulan quality flow dari fungsi grading dan risk matrix.

S19 / Pascapanen dan hub

Packing Signature / Weekly

Visual allocation 6 × 5 m (concept) | Fungsi kemasan BP; ruang bersama usulan

Realistic finish study based on Blender concept; small image details may vary.

FUNCTION

Mengemas single retail atau dua buah per pengiriman Weekly pada alur bersih.

LAYOUT

Dua benches packing 2,4 m, protective materials, label station, timbang dan scan; demo single Signature serta dua-fruit Weekly. Pintu belakang pada model dipusatkan pada ruang bebas antarfurniture.

EQUIPMENT

Meja bersih, material pelindung, label printer, QR scan, scales bila perlu.

CONNECTIONS

S17 release + S21 kemasan → S19 → S22/S23.

SERVICES

Power printer/scale; clean stock dari S21; dry task area.

FINISHES

Permukaan halus yang dapat dibersihkan, matte light-gray; clean bays dipisahkan dari returns. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

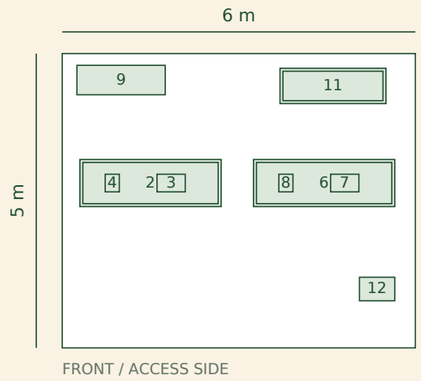
DESIGN REVIEW

30 m² konsep. Weekly dua buah per delivery, min empat minggu; jumlah meja belum dibuktikan laju packing.

SOURCE / STATUS

Bab IV Tabel 4.21; packaging Bab V; layout room = usulan.

DIMENSIONED CONCEPT



1. Packing Bench top · 2.4 × 0.8 m
2. Packing Bench shelf · 2.3 × 0.7 m
3. Weekly base · 0.48 × 0.3 m
4. Signature base · 0.24 × 0.3 m
5. Packing Bench top.001 · 2.4 × 0.8 m
6. Packing Bench shelf.001 · 2.3 × 0.7 m
7. Weekly base.001 · 0.48 × 0.3 m
8. Signature base.001 · 0.24 × 0.3 m

Full numbered equipment legend is in the interactive atlas.

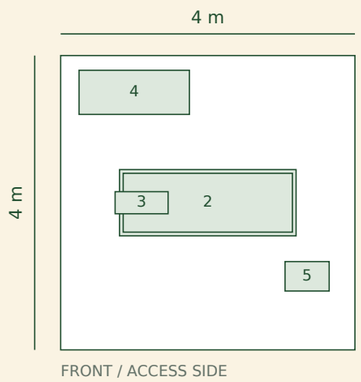
S20 / Pascapanen dan hub

Assembly Gift Edition

Visual allocation 4 × 4 m (concept) | Fungsi gift BP; zona usulan

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



1. Gift Assembly top · 2.4 × 0.9 m
2. Gift Assembly shelf · 2.3 × 0.8 m
3. Gift Second base · 0.72 × 0.3 m
4. Gift Inlay Stock shelf · 1.5 × 0.6 m
5. Finished Packs crate base.001 · 0.6 × 0.4 m

FUNCTION

Seleksi dan perakitan tiga buah Grade A+ dalam rigid box.

LAYOUT

Assembly bench, tiga-fruit rigid gift box, inlay/card station dan finish inspection; materials tersimpan tertutup.

EQUIPMENT

Meja assembly, inlay, kartu narasi, packaging QC.

CONNECTIONS

S17 seleksi A+ + S21 → S20 → S22/S23.

SERVICES

Task light; barcode/batch verification sebelum masuk cold chain.

FINISHES

Permukaan halus yang dapat dibersihkan, matte light-gray; clean bays dipisahkan dari returns. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

16 m² konsep, dapat berbagi ruang packing. Tidak mengubah Gift Edition menjadi dua buah.

SOURCE / STATUS

Bab IV Tabel 4.20/4.21; ruang assembly = usulan.

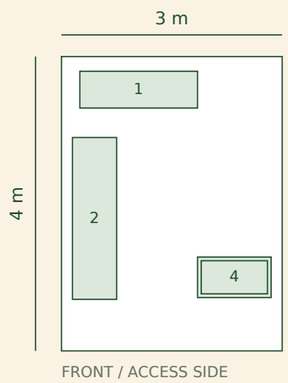
S21 / Pascapanen dan hub

Gudang kemasan bersih

Visual allocation 3 × 4 m (concept) | Usulan fasilitas bersama

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



1. Storage shelf.008 · 1.6 × 0.5 m
2. Side Storage shelf.008 · 0.6 × 2.2 m
3. Inventory Desk top.002 · 1 × 0.55 m
4. Inventory Desk shelf.002 · 0.9 × 0.45 m

FUNCTION

Menjaga box, inlay, labels dan kartu tetap bersih, kering dan siap dipakai.

LAYOUT

Rak box datar, inlay dan label stock tertutup; FIFO, off-floor storage dan packaging inspection bench.

EQUIPMENT

Rak tertutup/covered stock, FIFO lot packaging.

CONNECTIONS

S01 penerimaan kemasan melalui jalur bersih → S21 → S19/S20.

SERVICES

Dry storage conditions; light.

FINISHES

Permukaan halus yang dapat dibersihkan, matte light-gray; clean bays dipisahkan dari returns. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

12 m² konsep. Stock peak dan carton dimensions menentukan rak; tidak berbagi chemical store.

SOURCE / STATUS

Pemasok packaging Bab V; ruang = usulan.

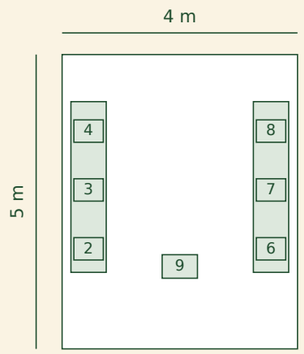
S22 / Pascapanen dan hub

Cold room pusat / staging dingin

Visual allocation 4 × 5 m (concept) | Cold storage pusat disebut workbook

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



1. Cold Rack shelf · 0.6 × 2.9 m
2. Cold Crate base · 0.5 × 0.38 m
3. Cold Crate base.001 · 0.5 × 0.38 m
4. Cold Crate base.002 · 0.5 × 0.38 m
5. Cold Rack shelf.004 · 0.6 × 2.9 m
6. Cold Crate base.012 · 0.5 × 0.38 m
7. Cold Crate base.013 · 0.5 × 0.38 m
8. Cold Crate base.014 · 0.5 × 0.38 m

Full numbered equipment legend is in the interactive atlas.

FUNCTION

Menjaga mutu sebelum distribusi sesuai proses yang divalidasi. Initial cold room ditempatkan pada HQ/central facility.

LAYOUT

Insulated room, strip curtain, accessible aisle, rack crates dan evaporator service position; external condenser bukan di dalam room. Pintu belakang pada model dipusatkan pada ruang bebas antarfurniture.

EQUIPMENT

Workbook Y0: cold storage/cold chain Rp 450 juta; temperature logging, racks, akses servis.

CONNECTIONS

S19/S20 → S22 → S23; site satelit memakai transfer/cold-box sesuai SLA.

SERVICES

Cold room vendor, temperature logging, defrost/condensate drain dan power.

FINISHES

Permukaan halus yang dapat dibersihkan, matte light-gray; clean bays dipisahkan dari returns. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

20 m² internal-envelope studi, bukan tonase final. Tidak ada setpoint 4–8°C yang dijadikan final atau angka di display. Satu unit pusat Y0 dari workbook.

SOURCE / STATUS

CAPEX_Detail B15/N15: 1 unit @ HQ Y0; additions Y4/Y5. Ukuran = TBD.

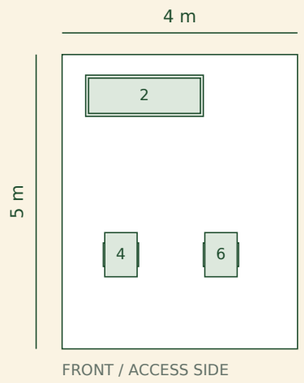
S23 / Pascapanen dan hub

Dispatch / handoff cold chain

Visual allocation 4 × 5 m (concept) | Fungsi logistik BP; ruang usulan

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



1. Order Check top · 2 × 0.7 m
2. Order Check shelf · 1.9 × 0.6 m
3. Outbound crate base · 0.6 × 0.4 m
4. Cold Box · 0.55 × 0.75 m
5. Outbound crate base.001 · 0.6 × 0.4 m
6. Cold Box.001 · 0.55 × 0.75 m

FUNCTION

Mencocokkan order, batch, pack, waktu dan kendaraan sebelum pengiriman.

LAYOUT

Order check bench, protected cold-box staging dan outbound trolley; finished packs tidak bertemu returns.

EQUIPMENT

Order check, cold box/reefer sesuai armada, trolley, timestamp handoff.

CONNECTIONS

S22/S19 → S23 → S26 → pelanggan/mitra.

SERVICES

Dispatch timestamps; cold-chain handoff sesuai SLA kanal.

FINISHES

Permukaan halus yang dapat dibersihkan, matte light-gray; clean bays dipisahkan dari returns. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

20 m² konsep. Jangan menahan cold product terlalu lama di staging suhu lingkungan.

SOURCE / STATUS

Logistik Bab V; CAPEX_Detail B16: 2 unit cool box motor pada Y0.

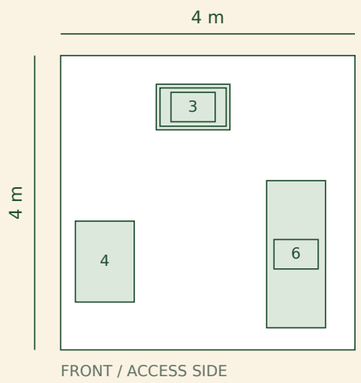
S24 / Pascapanen dan hub

Cuci crate / alat dan pengeringan

Visual allocation 4 × 4 m (concept) | Usulan zona basah terpisah

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



1. Wash Trough top · 1 × 0.62 m
2. Wash Trough shelf · 0.9 × 0.52 m
3. Wash Trough basin · 0.6 × 0.4 m
4. Dirty Incoming shelf · 0.8 × 1.1 m
5. Clean Drying shelf · 0.8 × 2 m
6. Drying Crate base · 0.6 × 0.4 m

FUNCTION

Mencuci dan mengeringkan reusable crates/tools, tanpa membasahi packing bersih.

LAYOUT

Wash trough, dirty incoming rack, clean drying rack dan splash separation; alur kotor → cuci → kering → cleanhold. Rak pengering ditarik ke depan agar pendekatan pintu belakang bebas.

EQUIPMENT

Wash point, rack drying, tools sesuai SOP, drain yang dirancang.

CONNECTIONS

Dirty return → S24 → dry clean hold → S16/G03.

SERVICES

Water, wastewater dan drain treatment sesuai gedung; hanging hose terparkir.

FINISHES

Lantai antiselip warna abu muda dengan sambungan bersih dan drain terkendali; detail kemiringan belum final. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

16 m² konsep. Bukan pencucian buah otomatis; chemicals, dwell dan rinse mengikuti SOP.

SOURCE / STATUS

Usulan flow kebersihan untuk crate/alat yang disebut CAPEX.

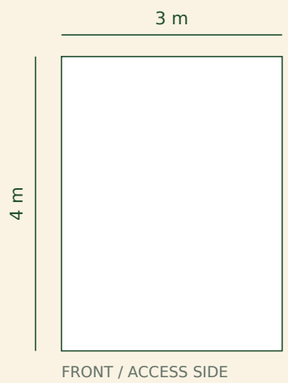
S25 / Pascapanen dan hub

Limbah tanaman / packaging / residual

Visual allocation 3 × 4 m (concept) | Usulan area terkontrol

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



FUNCTION

Mengumpulkan sisa tanaman, reject dan packaging; memisahkan limbah nutrisi/kimia sesuai proses.

LAYOUT

Covered bins untuk crop residue, packaging dan residual terpisah; trolley pickup; tidak melalui packing bersih.

EQUIPMENT

Bin tertutup dan terlabel, route removal, tempat tahan residual sesuai bahan.

CONNECTIONS

G04/S18/S24 → S25 → pengelola disposal site.

SERVICES

Controlled runoff dan removal route; ventilasi yard.

FINISHES

Lantai antiselip warna abu muda dengan sambungan bersih dan drain terkendali; detail kemiringan belum final. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

12 m² konsep. Jenis residu dan aturan disposal mengikuti bahan aktual.

SOURCE / STATUS

Risiko penyakit/efficiency BP; rincian waste area = usulan.

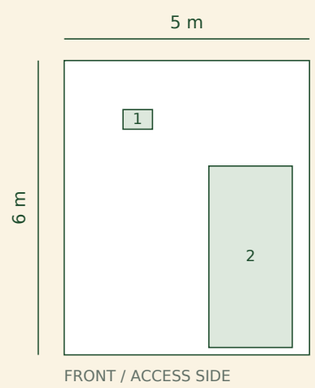
S26 / Site dan logistik

Lift barang / dock / transfer ke jalan

Visual allocation 5 × 6 m (concept) | Usulan integrasi fasilitas gedung

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



1. Dock Trolley crate base · 0.6 × 0.4 m
2. Vehicle Envelope · 1.7 × 3.7 m

FUNCTION

Menghubungkan rooftop dengan receiving/dispatch dan kendaraan di tingkat jalan.

LAYOUT

Lift door, landing trolley dan ground-level loading canopy; van cold-chain sebagai equipment envelope konsep.

EQUIPMENT

Lift/hoist yang disetujui gedung, loading point, trolley, jadwal lift.

CONNECTIONS

S01 input dan S23 buah punya jadwal/jalur terpisah; akses kendaraan di ground.

SERVICES

Building lift/dock infrastructure, akses servis dan scheduler.

FINISHES

Matte light-gray dry floor; threshold rata; furniture mudah dibersihkan. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

30 m² vignette. Lift payload, car dimensions dan dock height belum tersedia; tidak memvisualkan hoist yang sudah disetujui.

SOURCE / STATUS

Rooftop + distribution BP; akses vertikal belum terinci.

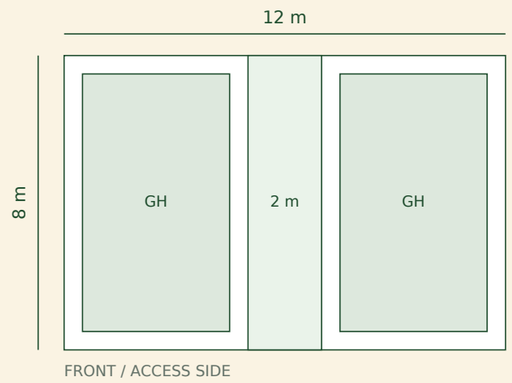
S27 / Site dan logistik

Sirkulasi site / egress / drain atap

Visual allocation 12 × 8 m (concept) | Usulan integrasi struktur gedung

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



FUNCTION

Jalur staf/trolley, akses service, jalur keluar dan drain roof yang tidak tertutup equipment.

LAYOUT

Dua tepi GH terlihat, dedicated walkway, roof drains dan edge guard; rute tidak menumpuk pallets pada waterproofing.

EQUIPMENT

Clear routes, outlet roof drain, edge protection sesuai desain gedung.

CONNECTIONS

Antar-GH ↔ shared rooms ↔ lift/stairs; drain gedung.

SERVICES

Roof drainage dan protection sesuai gedung.

FINISHES

Permukaan existing roof dilindungi sesuai gedung; jalur abu muda dan planter/DFT support pada skid terpisah. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

Area vignette 12 × 8 m, bukan plot site. Jalur utama 2 m konsep; egress, setbacks, guard dan roof loads tetap keputusan engineer.

SOURCE / STATUS

Renovasi rooftop CAPEX B19; detail engineering belum ada.

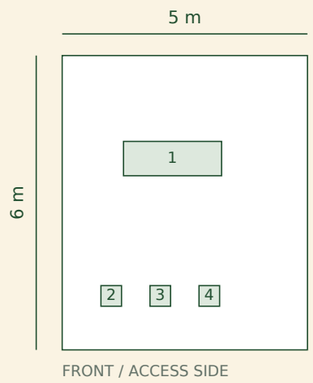
S28 / Pengembangan opsional

Open Farm reception / viewing

Visual allocation 5 × 6 m (concept) | Opsional; touchpoint BP, ruang usulan

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



1. Reception Desk · 2 × 0.7 m
2. Visitor Seat seat · 0.42 × 0.42 m
3. Visitor Seat seat.001 · 0.42 × 0.42 m
4. Visitor Seat seat.002 · 0.42 × 0.42 m

FUNCTION

Menerima dan mengedukasi pengunjung tanpa menjadikan semua jalur budidaya jalur wisata.

LAYOUT

Check-in counter, briefing zone, viewing barrier dan akses yang menghadap GH; tidak melintasi utilities/packing.

EQUIPMENT

Visitor check-in, viewing route, briefing/signage; hygiene jika masuk terbatas.

CONNECTIONS

Entrance publik → S28 → observation route → keluar; tidak lewat chemical/packing.

SERVICES

Booking/check-in; daylight; signage layer asli pada tahap produksi.

FINISHES

Matte light-gray dry floor; threshold rata; furniture mudah dibersihkan. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

30 m² konsep opsional. Bukan tiket/kegiatan yang telah dijual, dan tidak mengorbankan lintasan panen.

SOURCE / STATUS

Bab IV Brand Touchpoints: Open Farm; layout program = usulan.

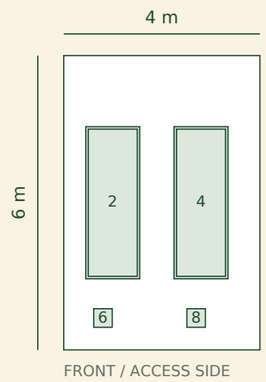
S29 / Pengembangan opsional

Pilot R&D / trial bay

Visual allocation 4 × 6 m (concept) | Opsional; program trial di panduan

Realistic finish study based on Blender concept; small image details may vary.

DIMENSIONED CONCEPT



1. Trial A top · 1.1 × 3.1 m
2. Trial A shelf · 1 × 3 m
3. Trial B top · 1.1 × 3.1 m
4. Trial B shelf · 1 × 3 m
5. Trial Loop A vessel · 0.36 × 0.36 m
6. Trial Loop A base · 0.378 × 0.378 m
7. Trial Loop B vessel · 0.36 × 0.36 m
8. Trial Loop B base · 0.378 × 0.378 m

FUNCTION

Trial varietas/proses pada area yang dibedakan dari produksi komersial.

LAYOUT

Dua trial benches dengan independent mini loops dan reference crops; batch trial terpisah dari release komersial.

EQUIPMENT

Label trial, loop/records terpisah, bench/mini bay sesuai trial.

CONNECTIONS

Agronomi → S29; output belum otomatis masuk pack premium.

SERVICES

Monitoring trial, sampling, cleaning dan experiment log.

FINISHES

Permukaan existing roof dilindungi sesuai gedung; jalur abu muda dan planter/DFT support pada skid terpisah. Off-white washable surfaces; galvanised steel on crop frames; forest green only on cabinets, crate, trim and identity details. Stainless steel in fruit/wet areas; sealed dry worktops in staff/admin zones.

DESIGN REVIEW

24 m² konsep opsional. Bila memakai commercial GH, kurangi kapasitas model; model ini tidak otomatis menambah GH ke roadmap.

SOURCE / STATUS

Panduan Bab V Improvement: R&D; area/kapasitas = belum final.

Sources, model and design decisions

PRIMARY PROJECT EVIDENCE

Latest BP: Melooloo_AfterSeminar2_BabIV_rev_28Sep2026.docx, Bab V infrastructure, DFT, layout and operations. Bab V chapter guide: Panduan_Bab_5_Strategi_Operasi.docx. Financial parameters: BAB8_FINAL_29Jul2026.xlsx computed parameter sheets. Preceding greenhouse infrastructure and product visualization guidelines dated 28 September 2026.

REFERENCE CONVENTION

Bracketed paragraph references in section pages are extracted paragraph indices used by the preceding source analysis, not Word page numbers. The functions grounded in the BP are identified separately from proposed geometry, furniture and optional areas.

TECHNICAL CROSS-CHECKS

CDC: https://www.cdc.gov/niosh/carbon-monoxide/about/index.html . Penn State Extension: https://extension.psu.edu/hydroponics-systems-nutrient-solution-programs-and-recipes . Blender render engine: https://developer.blender.org/docs/features/eevee/ .

EDITABLE GEOMETRY

The Blender source contains 31 scenes and 42 cameras, including the module and shared section scenes. It has 420 named planting anchors. Built, revised, rendered and saved inside Blender through computer use. Native renders are provided alongside the realistic finish studies.

COMPLETE HANDOFF

Open index.html for searchable section navigation, realistic/Blender/plan switches and equipment legends. The package also includes all 42 realistic images, 42 native renders, 41 SVG plans, section schedule, full design data, Blender source and reproducible construction scripts.

REMAINING DESIGN RECONCILIATION

Net planting area; final nutrient tank BOM and hydraulics; canopy and crop access; actual roof/site geometry and structural load; vendor dimensions and maintenance clearances; cold room throughput and setpoints; sanitation, drainage and egress. These are proposed design studies, not as-built or construction drawings.